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1  /// strings :
2  /*
3
4  ~ a string :
5      - variable contains a collection of characters
6      |   - surrounded by double quotes " "
7      - strings is Array of characters
8      - string is a object of the string class in C++
9      - string class is a part of the C++ Standard Library
10
11 ~ include the string library :
12     - to use the string class, we need to include the string library
13     #include <string>
14
15 ~ array :
16     - collection of items of the same type
17     - fixed size
18     - stored in contiguous memory locations
19 # to get the specific item in the array, we can use the index number of the item 🐼
20     - index starts from 0
21     variable_name[index_number]
22     - example :
23         string my_string = "Hello, World!";
24         char first_character = my_string[0]; // first_character will be 'H'
25         char second_character = my_string[1]; // second_character will be 'e'
26         char third_character = my_string[2]; // third_character will be 'l'
27         char fourth_character = my_string[3]; // fourth_character will be 'l'
28         char fifth_character = my_string[4]; // fifth_character will be 'o'
29         char sixth_character = my_string[5]; // sixth_character will be ','
30
31 ~ methods :
32     * length() :
33         returns the length of the string
34         string my_string = "Hello, World!";
35         int length = my_string.length(); // length will be 13
36     * getline
37         to read a string with spaces
38         syntax :
39         getline(cin, variable_name);
40         example :
41         string my_string;
42         cout << "Enter a string: ";
43         getline(cin, my_string);
44         cout << "You entered: " << my_string << endl;
45
46 ~ concatenation :
47     - joining two strings together
48     - we can use the + operator to concatenate two strings
49     string s1 = "Hello, ";
50     string s2 = "World!";
51     string s3 = s1 + s2; // s3 will be "Hello, World!"
52
53 ~ note:
54     ^ when use stoi function to convert string to integer,
55     ^ the string should contain only digits, otherwise it will throw an exception.
56

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57      ! read string with spaces :
58      - to read a string with spaces, we can use the getline() function
59      - syntax :
60      | getline(cin, variable_name);| 🦆
61      - example :
62      | string my_string;
63      | cout << "Enter a string: ";
64      | getline(cin, my_string);
65      | cout << "You entered: " << my_string << endl;
66
67  */
68
69
70  //~ example :
71

```

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72  #include <iostream>
73  #include <string>
74  using namespace std;
75
76  int main()
77  {
78      string my_string = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
79      cout << "The length of the string is: " << my_string.length() << endl;
80
81      cout << my_string[0] << endl; // A
82      cout << my_string[6] << endl; // G
83      cout << my_string[12] << endl; // M
84
85
86      string s1="10" , s2="20";
87      string s3 = s1 + s2; // concatenation
88      cout << s3 << endl; // 1020
89
90
91      int sum = stoi(s1) + stoi(s2); // convert string to integer and then add
92      cout << "The sum is: " << sum << endl; // The sum is: 30
93
94      string fullname;
95      cout << "Enter your full name: ";
96      getline(cin, fullname); // read string with spaces
97      cout << "You entered: " << fullname << endl;
98      return 0;
99  }

```